

Clinical Risk Assessment Report: Patient 57

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 11.0% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.50x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Patient Age (Age), which reduced the patient's absolute risk by 1.3%. Biologically, this feature patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

2. Key Pathological & Imaging Drivers (Elevating Risk)

No major risk-elevating features observed in the top drivers.

3. Protective / Mitigating Factors (Reducing Risk)

Patient Age (Age) **Value: 78.0 [Top 4%] (-1.3%)**

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.16 [Avg (59th %ile)] (-1.2%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 0.2 [Top 22%] (-0.8%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Carcinoembryonic Antigen (CEA) **Value: 1.3 [Bottom 10%] (-0.7%)**

Elevated levels are strongly associated with increased tumor burden and potential for micrometastasis.

Tumor Sphericity (PET) (SHAPE_Sphericity(onlyFor3DROI)).PET) **Value: -1.77 [Bottom 6%] (-0.6%)**

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Tumor Pixel Correlation (CT) (GLCM_Correlation.CT) **Value: -0.0 [Avg (46th %ile)] (-0.6%)**

Measures the linear dependency of gray levels; atypical values corroborate a complex, non-uniform spatial cellular arrangement.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.11 [Avg (59th %ile)] (-0.5%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: -0.1 [Avg (41th %ile)] (-0.5%)**

Reflects the most dense solid components or calcifications within the primary tumor lesion.

Machine Learning Feature Attribution (SHAP Decision Path)

